



$$\mathbf{s}_t = \mathbf{q}_t \mathbf{K}_t^\top = [\mathbf{q}_t \mathbf{K}_{<t}^\top \quad \mathbf{q}_t \mathbf{k}_t^\top], \quad \mathbf{o}_t = \sum_{i=1}^t \alpha_{t,i} \mathbf{v}_i$$